higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

COMMUNICATION-ELECTRONICS N5

28 MARCH 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 The ability of a tuned circuit to discriminate against frequencies other than its resonance frequency (3)

1.2 Q = 100; R = 100 Ω; E = 10 mV; $f = 455$ kHz; I = 0,1 A

1.2.1
$$V_L = EQ$$
$$V_L = 10\,mV \times 100$$
$$V_L = 1\,V$$

$$V_C = V_L = 1\,V$$

(5)

1.2.2
$$\frac{X_L}{R} = 100$$
$$X_L = 100 \times R$$
$$L = \frac{100(100\,\Omega)}{2\pi f}$$
$$L = \frac{1\ \ 10\,000\ \Omega}{2 \times \pi (455 \times 10^3\,Hz)}$$
$$L = 3{,}5\,\mu H$$

(3)

1.3 1.3.1
$$Z_{RL} = R + jX_L$$
$$Z_{RL} = 3 + j4$$
$$Z_{RL} = 5\Omega \angle 25.32°$$

$$Z_C = 0 - jX_C$$
$$Z_C = 0 - j3$$
$$Z_C = 3\Omega \angle -90°$$

$$I_{LR} = \left(\frac{Z_C}{Z_{LR} + Z_C}\right) I_S$$
$$I_{LR} = \left(\frac{3\Omega \angle -90°}{3 + j4 - j3}\right) 50\,A \angle 30°$$
$$I_{LR} = 75\,A \angle -41{,}565°$$

(6)

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1.3.2

$$I_C=\left(\frac{Z_{LR}}{Z_{LR}+Z_C}\right)I_S$$
$$I_C=\left(\frac{5\Omega\angle 25.32°}{3+j4-j3}\right)50A\angle 30°$$
$$I_C=\left(\frac{100V\angle 55,32°}{2\Omega\angle 18,433°}\right)$$
$$I_C=50A\angle 36,89°$$

(5)
[22]

QUESTION 2

2.1

$$Q_P=\frac{X_{L_P}}{R_1+Ri};\ f_O=\frac{1}{2\times\pi\sqrt{LC}}=\frac{1}{2\times\pi\sqrt{(120\times10^{-6}H)(500\times10^{-12}F)}}=649,747kHz$$

$$Q_P=\frac{2\times\pi\times f\times L_i}{R_1+Ri}$$

$$Q_P=\frac{2\times\pi(649,747\times10^3Hz)(120\times10^{-6}H)}{4\Omega}$$

(3)

$$Q_P=122,475$$

$$Q_S=\frac{X_L}{R_S}$$

$$Q_S=\frac{2\times\pi\times f\times L_l}{R_2+R_l}$$
$$Q_S=\frac{2\times\pi(649,747\times10^3Hz)(500\times10^{-6}H)}{30\Omega}$$
$$Q_S=68,04$$

(3)

2.2

$$k_c=\frac{1}{\sqrt{Q_PQ_S}}$$
$$k_c=\frac{1}{\sqrt{122.47\times68.04}}$$
$$k_c=0,11$$

(2)

2.3 $$k_c = \frac{m}{\sqrt{L_P L_S}}$$

$$k_c = \frac{7,35\times10^{-6}\,H}{\sqrt{(120\times10^{-6})(500\times10^{-6})}}$$

$$k_c = 0,03$$

(2)
[10]

QUESTION 3

3.1 $$L = \frac{R_O}{\pi f_C};$$

$$= \frac{600\,\Omega}{\pi(4.3\times10^3\,Hz)}$$

$$= \frac{1}{600(\pi)(4,3\times10^3\,H_z)}$$

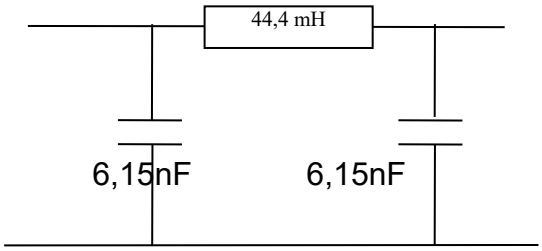
$$= 44,4\,mH$$

$$C = \frac{1}{R_O \pi f_c}$$

$$= 12,3\,nF$$

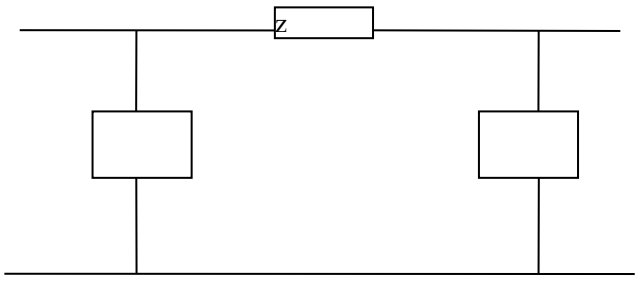
$$\frac{C}{2} = 6,15\,nF$$

T – Type/Tipe



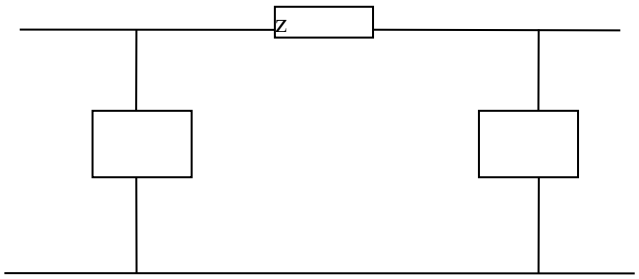
π-TYPE LOW PASS FILTER (12)

3.2 3.2.1



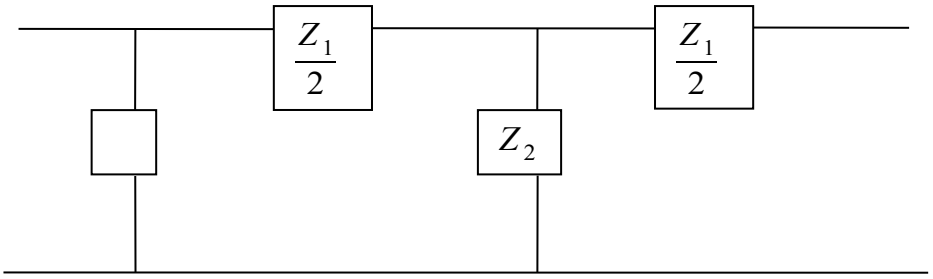
$$Z_1 = j\omega L;$$ $$Z_2 = \frac{1}{j\omega C}$$

3.2.2



$$Z_1 = \frac{1}{j\omega C}; \qquad Z_2 = j\omega L$$

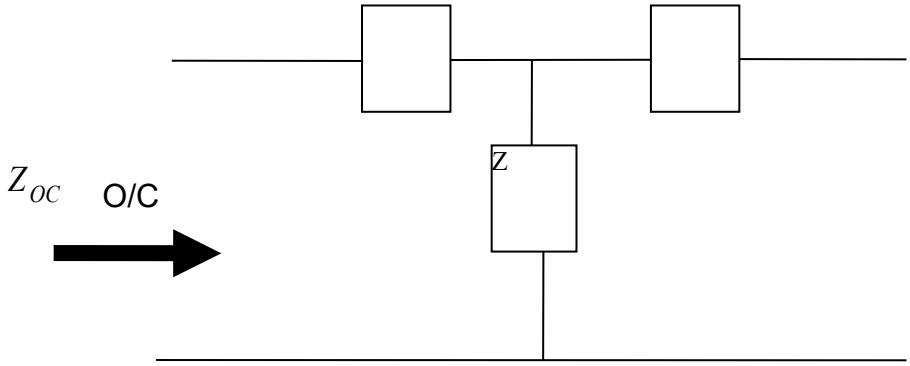
3.2.3



$$Z_1 = j\omega L; \qquad Z_2 = \frac{1}{j\omega C}$$

(3 × 4) (12)
[24]

QUESTION 4



$$Z_{OC}=\frac{Z_1}{2}+Z_2 \quad \text{.....1}$$

From 1

$$Z_{OC}^2=\frac{Z_1^2}{4}+Z_1Z_2+Z_2^2$$

$$Z_2^2=Z_{OC}^2-\left(\frac{Z_1^2}{2}+Z_1Z_2\right)$$

$$Z_2^2=Z_{OC}^2-Z_{OT}^2$$

$$Z_2^2=Z_{OC}^2-\left(\sqrt{\frac{Z_1^2}{4}+Z_1Z_2}\right)^2$$

$$Z_2^2=Z_{OC}^2-\left(\sqrt{Z_{OC}\times Z_{SC}}\right)^2$$

$$Z_2^2=Z_{OC}^2-\left(Z_{OC}\times Z_{SC}\right)$$

$$Z_2=\sqrt{Z_{OC}\left(Z_{OC}-Z_{SC}\right)}$$

$$Z_2=\sqrt{400\left(400-300\right)}$$

$$Z_2=200\,\Omega$$

$$\frac{Z_1}{2}=Z_{OC}-Z_2$$

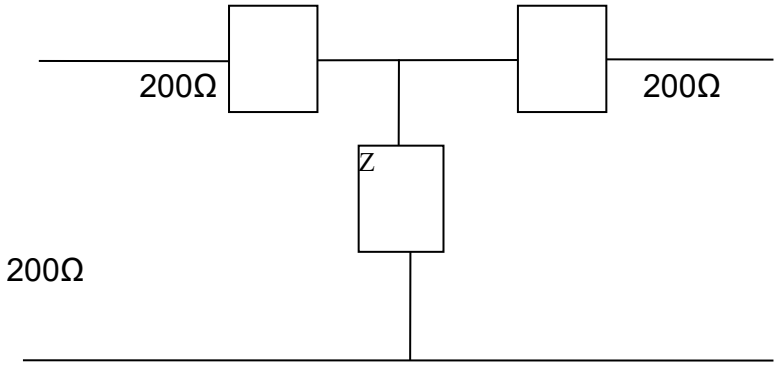
$$Z_1=2\left[Z_{OC}-Z_2\right]$$

$$Z_1=2\left[Z_{OC}-\sqrt{Z_{OC}\left(Z_{OC}-Z_{SC}\right)}\right]$$

$$Z_1=2\left[400-\sqrt{400\left(400-300\right)}\right]$$

$$Z_1=2\left(400-200\right)$$

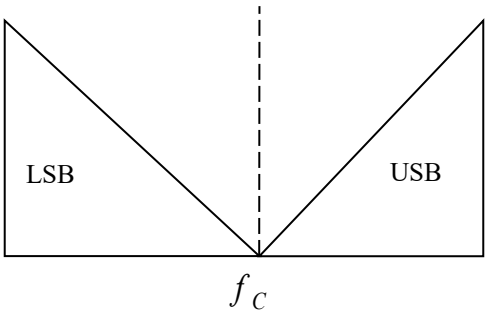
$$Z_1=400\,\Omega$$



[15]

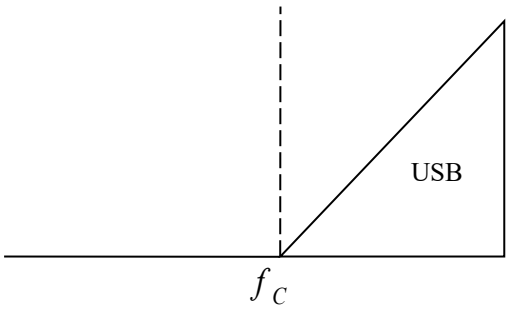
QUESTION 5

5.1



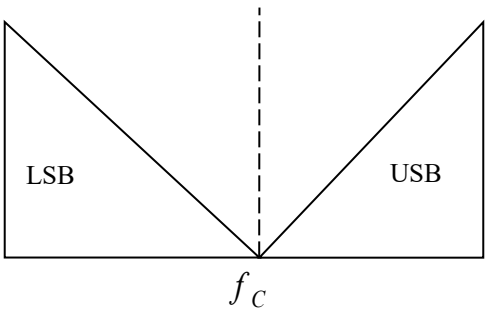
(3)

5.2



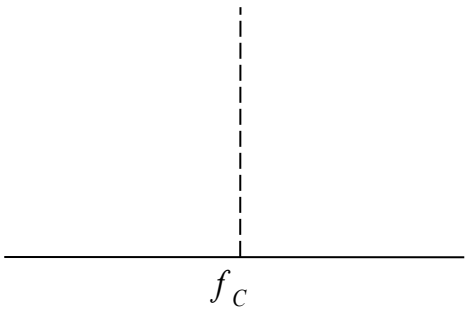
(1)

5.3



(2)

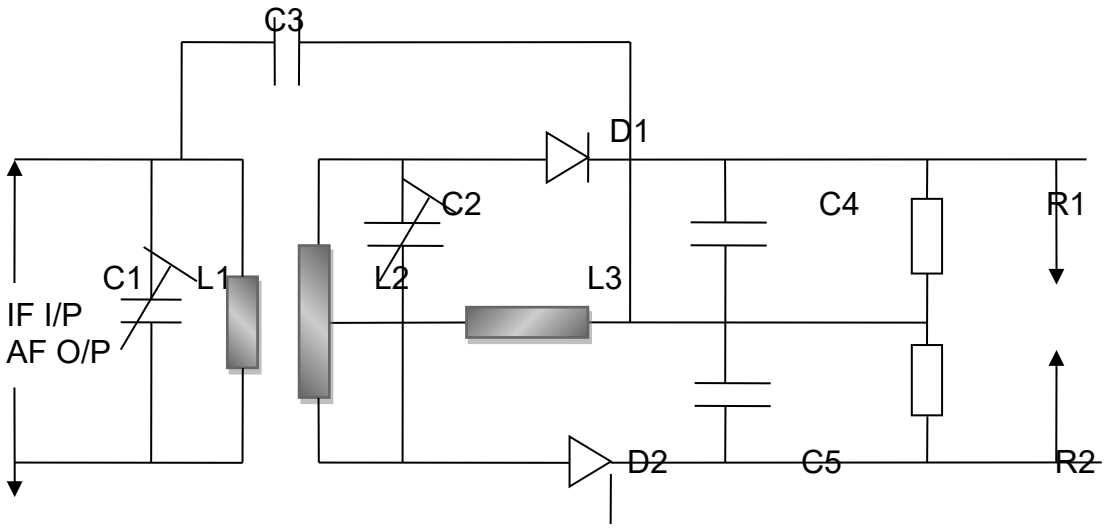
5.4



(1)
[7]

QUESTION 6

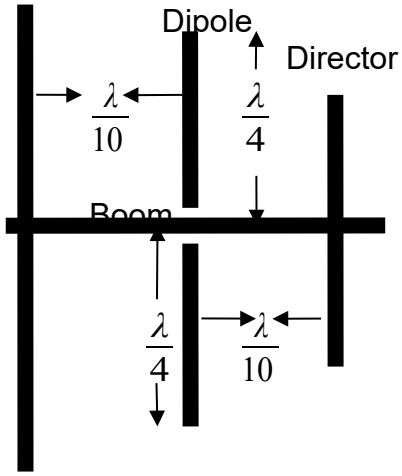
The discriminator/ratio detector/circuit variations.



CIRCUIT VARIATIONS EXIST **[12]**

QUESTION 7

7.1 Reflector



OR VARIATIONS (6)

7.2 7.2.1 When the source impedance equals the load impedance

7.2.2 An antenna whose length is a multiple or submultiple of the wavelength

(2 × 2) (4)

[10]

TOTAL: 100

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